Week 15 Capstone — Client Brief (Template + Worked Example)

Your team writes its own client brief for the dataset you chose — there's no instructor-provided brief this time. Write it **before** starting the in-class exercise notebook; it sets the context for your executive summary memo (Section E) and your Capstone Report.

A good brief has three parts: a named client/stakeholder, a short paragraph establishing why they care about your data, and 2–3 specific questions they want answered.

Write the questions your client would actually ask, not the ones your methods are shaped to answer. You will run all four methods either way — regression, hypothesis test, Monte Carlo, bootstrap — and part of the work is reporting which of them spoke to your client's question and which had nothing to say about it. An engineer runs more analysis than they report; deciding what earned a place in the briefing is the judgement being assessed, and you cannot exercise it if the questions were reverse-engineered from the method list to begin with.

Template

Client: [A named, plausible stakeholder — a real job title at a real or realistic organization]

[2–4 sentences establishing the client's situation: what they're responsible for, why this data matters to them, and what decision or budget question is riding on the answer. End with 2–3 numbered questions the client wants answered — write these to be answerable by your regression, hypothesis test, and Monte Carlo results.]

Your dataset: [filename or source] — [row count] observations, [what each row represents], [time period/location if relevant].

Worked Example — Hospital (using a legacy building-energy dataset)

Client: Facility Manager, Banner University Medical Center (hypothetical 800-bed acute-care hospital, Tucson)

Your hospital runs 24 hours a day, 365 days a year. Cooling is not optional — patient rooms, operating theaters, and pharmacy storage all have strict temperature requirements. Last fiscal year, cooling electricity accounted for 38% of the facility's total energy bill. The CFO has asked for a data-driven analysis to answer three questions: (1) How strongly does outdoor temperature drive cooling load? (2) Is there a measurable difference in cooling during standard business hours versus overnight? If so, what does the coefficient mean operationally for a 24/7 facility? (3) What is the expected annual cooling cost, and what is the realistic range if electricity rates fluctuate?

Your dataset: `Week15_Hospital_Dataset.csv` — 8,760 hourly observations, Tucson climate, 2023. Predictor (X) = outdoor temperature; outcome (Y) = cooling energy; group = business hours vs. non-business hours.

Tips

- **Pick a client who would realistically pay for this analysis.** "A city engineer deciding whether to add a signal at an intersection" is more specific and more gradeable than "someone interested in traffic."
- **Keep the question answerable from data — any data.** The test is not whether it matches a method you were taught; it is whether a number could settle it. "Should we build a new facility?" fails that test no matter what you run, because it turns on budget, politics and land. "Is the overnight shift costing us more per unit than the day shift, and by how much?" passes: it is a real question a real client has, and data can answer it.
- **If your grouping variable produces a counterintuitive result** (e.g., a hotel dataset where evening peak occupancy makes daytime "business hours" look like a *lower*-cooling period), don't avoid it. Write a question that specifically asks your team to explain it. That's usually the most interesting part of the analysis.